

Genome-wide identification of the YABBY gene family and functional characterization of TaYABBY4A in wheat (*Triticum aestivum* L.)

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Abstract

Background

YABBYs are plant-specific transcription factors that play crucial roles in plant growth, development, and stress responses. Despite extensive studies in various plant species, a systematic analysis of *YABBYs* in wheat grains is still lacking.

Results

In this study, 21 *TaYABBYs* were identified using the Chinese Spring wheat genome database and were divided into five subfamilies through phylogenetic analysis. Gene collinearity analysis revealed the evolutionary characteristics of the *TaYABBYs*. Analysis of cis-acting elements in the promoter region identified elements related to endosperm development. SNP analysis uncovered genetic variations within the *YABBY* gene family. Meanwhile, RNA-Seq and qRT-PCR techniques were employed to explore the expression patterns of *TaYABBYs*, and the results showed that these genes are differentially expressed in different wheat tissues. Additionally, we selected *TaYABBY4A* from the CRC subfamily for overexpression in *Arabidopsis thaliana* to verify the function of *TaYABBYs*. Overexpression of the *TaYABBY4A* in *Arabidopsis* resulted in delayed bolting and flowering, as well as reductions in the number and diameter of rosette leaves and seed size.

Conclusions

This study further confirms that the *YABBY* gene family plays an important regulatory role in the growth and development of wheat, providing a reference for in-depth exploration of the functions of *YABBYs* in wheat.

Introduction

Transcription factors (TFs) are key regulators of gene expression, either alone or combined with other factors, by binding to the cis-elements of their target genes. One gene can control an important agronomic trait. As a small family of transcription factors, the *YABBYs* are predominantly observed in seed plants.[1]. A feature of the members of this family is that they share two highly conserved domains: the classical C2C2 zinc finger domain in N-terminal region, and the basic helix-loop-helix (bHLH) in the C-terminal region, called the *YABBY* domain[2]. *YABBYs* can regulate gene expression as both activators and suppressors, and the functions of *YABBYs* correlate with their expression profiles directly or indirectly [3–5]. Studies have found that *YABBYs* function in plant development and growth[6, 7].

To date, researchers have identified the *YABBY* gene family in different plant species through genome-wide analysis[8]. A total of 6 *YABBYs* had been identified in *Arabidopsis thaliana* [5], 8 in *Oryza sativa* [9],

17 in *Glycine max* [10], 7 in *Vitis vinifera* L. [11], and 9 in *Solanum lycopersicum* L. [12]. During rice domestication, OsSh1 and ObSh3 played a key role in promoting seed shattering[13]. The FIL/YAB3 group genes (*OsYABBY3/4/5*) were highly transcribed in the spikelets, while *OsDL* was mainly expressed in pistils, inner and outer epidermis, and spikelets[13]. Overexpression of grape *VvYABBY4* in tomato was found to affect plant growth and development as well as fruit formation[14]. Orashakova discovered that *EcCRC* transcription was downregulated during ovule development[15]. Some identified YABBY homologous genes in cotton were expressed in ovules[16]. Overexpression of *SiDL* from the YABBY family of foxtail millet in Arabidopsis resulted in dwarf plants and reduced fruit and seed diameters[17]. These research results indicate that YABBYs play an important role in plant reproductive growth and development. Recent studies have shown that YABBYs have important regulatory functions in gramineous crops[18]. Wheat (*Triticum aestivum* L.) is a globally crucial food crop, with its grain development directly determining yield and quality[19]. TFs are pivotal in the complex regulatory network of wheat grain development[7]. However, their specific role in wheat grain development remains unclear.

In this study, a genome-wide search method utilizing the latest Chinese Spring wheat genome information was employed to identify 21 *TaYABBYs*. Through bioinformatics analysis, the physicochemical properties, phylogenetic relationships, gene structures, conserved motifs, chromosomal locations, and cis-acting promoter elements of these genes were systematically analyzed. Haplotype analysis revealed the genetic variation characteristics of *TaYABBYs* in wheat. Subsequently, based on publicly available transcriptome data and RNA-seq data of developing grains (grain filling stage) from Yaomai 36, Pinyu 8175, Pinyu 8155, and Yaomai 30, the expression patterns of *TaYABBYs* in different wheat tissues and endosperm development stages were analyzed. Finally, the function of *TaYABBY4A* was identified in *Arabidopsis thaliana*. This study aims to analyze wheat *YABBYs* and their functional characteristics at the genome-wide level, laying a foundation for in-depth exploration of their evolutionary history and functional mechanisms.

Results

Identification and sequence characteristics analysis of *TaYABBYs*

Using the Pfam and Phytozome v13 programs, 21 members of the YABBY gene family were identified in the Chinese Spring genome. Analysis of the amino acid sequences produced the sequence logo and annotation of the YABBY domain (Fig. 1). All *TaYABBYs* possess two conserved DNA-binding domains: an N-terminal C2-C2 zinc finger domain and a C-terminal YABBY domain. Within the C2-C2 zinc finger domain, the cysteine (C) and histidine (H) residues responsible for Zn²⁺ binding are conserved. At the C-terminus, within the YABBY domain, 27 amino acid residues are 100% conserved, including five alanine (A), three proline (P), two serine (S), and three isoleucine (I) residues. This conservation is essential for the domain's structure and function. Notably, *TaYABBY2A/B/D* and *TaYABBY4A/5B/5D* exhibit higher

variability in both domains, suggesting functional diversity. This divergence may equip these genes with unique roles in plant growth, development, and environmental responses.

According to the annotation of the wheat genome, among the 21 wheat chromosomes, 21 *TaYABBYs* were found to be unevenly distributed across 15 chromosomes (Table 1). Based on their consecutive chromosomal positions and homology relationships, these *YABBYs* were systematically designated as *TaYABBY1A-TaYABBY8D*. The coding sequence (CDS) lengths of the *TaYABBYs* varied significantly. The deduced length of CDS ranged from 495 bp (*TaYABBY2D*) to 894 bp (*TaYABBY1A/B*). Amino acid sequence lengths also varied widely, from 164 aa (*TaYABBY2D*) to 297 aa (*TaYABBY1A/B*), leading to substantial differences in predicted protein molecular weights, ranging from 17.76 kDa (*TaYABBY2D*) to 31.44 kDa (*TaYABBY1B*). Additionally, the theoretical isoelectric points (pI) of these proteins varied from 5.62 (*TaYABBY2A*) to 9.48 (*TaYABBY7B*). The subcellular localization of all 21 *TaYABBYs* was predicted by Plant-mPLOC, and the results indicated that these genes are situated in the nucleus.

Phylogenetic analysis, chromosomal location, gene structure, and conserved motifs of

Phylogenetic analysis, chromosomal location, gene structure, and conserved motifs of *TaYABBYs*
To investigate the evolutionary relationships between wheat and other plant species, a comprehensive phylogenetic tree was constructed using the Neighbor-Joining (NJ) method (Fig. 2A). Like *Arabidopsis*, *TaYABBYs* were categorized into five subfamilies: FIL/YAB3, YAB2, CRC, INO, and YAB5. However, the wheat *TaYABBYs* do not include the YAB5 subfamily. Within these, the YAB2 and FIL/YAB3 subfamilies had a larger number of members (with 6 in YAB2 and 9 in FIL/YAB3), while INO and CRC had the fewest, with 3 members each. The findings indicate that *YABBYs* in wheat are relatively conserved and may have acquired functional diversity throughout wheat evolution, potentially associated with its distinctive physiological and morphological characteristics. A total of 21 *TaYABBYs* were identified within the wheat genome, distributed unevenly across 15 of the 21 chromosomes (Fig. 2B).

To investigate the gene structure and conserved motifs of the *TaYABBYs*, we predicted exon-intron structures and ten conserved motifs using the MEME program and visualized them with TBtools (Fig. 2C). Analysis of 21 *TaYABBYs* revealed that all subfamilies exhibited similar exon-intron architectures. Furthermore, the gene structures of members within the same subfamily were more similar to each other than to those of members from different subfamilies. Specifically, members of the FIL/YAB3 and CRC sub-families typically possessed seven exons and six introns, whereas those of the YAB2 sub-family generally had six exons and five introns. However, *TaYABBY7D* in the YAB2 family was an exception, featuring seven exons and six introns, which may be attributed to evolutionary changes. Analysis of conserved motifs revealed that motifs 1, 2, 3, and 6 were present in every member, indicating that they are likely specific to the *YABBYs*. Further investigation indicated that motif 1 forms the *YABBY* domain at the C-terminus, and motifs 2 and 3 constitute the C2-C2 domain at the N-terminus. The FIL/YAB3 sub-family was characterized by motif 9. Notably, motif 10 was present in both the YAB3

and YAB2 clades, whereas TaYABBY3A lacks motif 2. The INO sub-family possessed a unique set of motifs (motifs 1, 2, 3, 5, 6), while members of the YAB2 (TaYABBY7A, TaYABBY7B, and TaYABBY7D) and CRC sub-families shared the same set of motifs (motifs 1, 2, 3, 4, 6, 7). The prediction of functional motifs indicated that members of each subgroup shared the same conserved motifs, suggesting that they may have similar functions.

Intraspecific and intermediate collinearity analysis of

Intraspecific and intermediate collinearity analysis of *TaYABBYs*

We utilized MCScanX within TBtools to examine tandem and fragment replication events in wheat (Figure S1). The findings indicated the presence of 24 pairs of fragment replication genes and 11 pairs of tandem replication genes within *TaYABBYs*. Additionally, it was observed that gene duplication events were more prevalent on chromosome 5 compared to the other chromosomes. To assess the evolutionary constraints on duplicated gene pairs, the Ka and Ks parameters were computed. All *TaYABBYs* homologous genes exhibited a Ka/Ks < 1, implying that they experienced strong purifying selection throughout their evolution (Table 2).

Furthermore, to elucidate the evolutionary mechanism of *TaYABBYs*, comparative syntenic blocks were constructed among Arabidopsis, wheat, and rice (Fig. 3). We identified 19 syntenic orthologous gene pairs between wheat and rice, where multiple *TaYABBYs* were matched with a single *OsYABBY*. In contrast, none were found between wheat and Arabidopsis, likely due to their early evolutionary divergence as a dicot and a monocot, respectively, hindering the identification of collinear gene regions.

Analysis of cis-acting elements in the promoter regions of

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To uncover the potential functions of *TaYABBYs*, 2000 bp upstream sequences of 21 *TaYABBYs* transcription start sites were extracted from the wheat genome using TBtools and analyzed for *cis*-acting elements via the online PlantCARE database (Fig. 4, Table S1). The predicted elements primarily relate to plant hormone, development, stress, and light responses. Notably, 18 seed-specific and 20 zein metabolism-regulatory elements, directly related to endosperm development, were detected. Additionally, 269 light-responsive, 253 plant hormone-responsive, and 73 stress-responsive elements were found, with light-responsive elements being prevalent in most *YABBY* promoters. These findings suggest that *TaYABBYs* may function in light and hormone responses, plant growth, development, and stress responses.

Haplotype analysis of

Haplotype analysis of *TaYABBYs*

To further explore the genetic variation of *TaYABBYs*, haplotype analysis was performed on 21 *TaYABBYs* with genetic variation information using Lufei resequencing datas (Table 3). The results showed that the number of haplotypes of *TaYABBYs* was lower in cultivated varieties, but higher in landrace varieties and Spelt. Globally, the number of haplotypes of *TaYABBYs* was higher in Asia and Europe, while the number of haplotypes in Africa, North America, and South America was lower. Concurrently, we also analyzed the haplotypes of spring wheat and winter wheat. The results showed that the number of haplotypes of spring wheat and winter wheat was greater than that of facultative wheat. Notely, there was no genetic variation of *TaYABBY1B/3A/2B/6A/8B/8D*, indicating that these genes were relatively stable and had better adaptability to different environments. In contrast, genes such as *TaYABBY5B/7A/6B/7B/7D/8A*, exhibit relatively high haplotype variation, suggesting greater genetic diversity. These distributions provide insights into the genetic variation and adaptation of *TaYABBYs* in different wheat-related groups.

Anaysis of *TaYABBYs* expression pattern based on RNA-seq

To investigate the functions of *TaYABBYs*, we analyzed the spatiotemporal expression of 21 *TaYABBYs* in different tissues and during various stages of endosperm development (Fig. 5A, Table S2). All genes exhibited low or negligible expression in roots and stems, but higher expression levels in spikes and endosperm. Genes within the same subfamily displayed both similarities and distinct differences in their expression patterns. During endosperm development, members of the YAB3/FIL and CRC subfamilies remained unexpressed. In contrast, *TaYABBY2D* from the INO subfamily, as well as *TaYABBY6A/B/D* and *TaYABBY7A/B* from the YAB2 subfamily, were expressed. *TaYABBY6A/B/D* and *TaYABBY7A/B* exhibited relatively high expression levels at 10 days post-pollination (10DPA) of endosperm development. *TaYABBY2D* and *TaYABBY6B/D* showed relatively higher expression levels at 20DPA, and only *TaYABBY2D* continued to be expressed at 30DPA. These findings suggest that *TaYABBYs* play a role in the process of reproductive growth, which aligns with the outcomes of promoter analysis.

Based on *TaYABBYs* predictions regarding wheat grain development, the transcriptomes of *TaYABBY* during the grain-filling stage were sequenced for four wheat varieties: 'Yaomai 36', 'Pinyu 8175', 'Pinyu 8155', and 'Yaomai' 30 (Fig. 5B and Table S3). The results indicated that the genes from the YAB2 and CRC subfamilies in all four varieties displayed higher expression at the milk ripe stage (MRS), moderate expression at the dough stage (DP), and lower expression at the wax ripe stage (WRS), with significant differences noted. In 'Yaomai 36', among the genes of the INO subfamily, only *TaYABBY2D* exhibited the highest expression level at the WRS stage; for genes of the YAB3/FIL subfamily, *TaYABBY3B*, *TaYABBY3D*, *TaYABBY3A*, *TaYABBY8A*, *TaYABBY8B*, and *TaYABBY8D* were highly expressed at the DP stage, whereas *TaYABBY1A*, *TaYABBY1B*, and *TaYABBY1D* were highly expressed at the MRS stage. In 'Pinyu 8175', the INO subfamily gene *TaYABBY2D* exhibited the lowest expression at the DP stage; among the YAB3/FIL subfamily genes, with the exception of *TaYABBY1B*, which was highly expressed at the MRS stage, the others were highly expressed at both the DP and WRS stages. In 'Pinyu 8155', the INO subfamily genes were highly expressed at the WRS stage and exhibited low expression at the MRS and DP stages; among the YAB3/FIL subfamily genes, only *TaYABBY3D* was highly expressed at the WRS

stage. In 'Yaomai 30', the INO subfamily gene *TaYABBY2D* exhibited the highest expression level at the MRS stage; among the YAB3/FIL subfamily genes, *TaYABBY3B*, *TaYABBY3D*, and *TaYABBY1B* were highly expressed at the MRS stage, while *TaYABBY8B* and *TaYABBY1D* were highly expressed at the DP stage. The results indicate that the *TaYABBYs* exhibit clear spatiotemporal specificity during grain development, with their expression levels varying significantly across different developmental stages. We chose *TaYABBY4A* from the CRC family, which is closely associated with grain development, for functional validation of the *TaYABBYs*.

Identification, subcellular localization, and transcriptional activity of

Identification, subcellular localization, and transcriptional activity of *TaYABBY4A*

To further validate the accuracy of transcriptome sequencing and verify the function of *TaYABBYs*, we conducted qRT-PCR detection of *TaYABBY4A* during the endosperm development of wheat grains. The results indicated that the expression trend was consistent with the RNA-seq data prediction, exhibiting a gradual decrease in expression levels across the three stages of grain development (Fig. 6A). Subsequently, real-time fluorescent quantitative PCR was performed on the 10 obtained transgenic *Arabidopsis* lines overexpressing *TaYABBY4A*. The results (Fig. 6B) indicated that the expression levels of OE1-OE10 were significantly higher than that of the WT, with OE3 and OE5 exhibiting the highest relative expression levels. Consequently, OE3 and OE5 were selected for further studies.

The full-length cDNA of *TaYABBY4A* was cloned into the pGBKT7 vector, and its self-activation ability was verified through yeast experiments (Fig. 6C). The negative control pGBKT7, positive control pGADT7-PtrWOX13A, and pGBKT7-*TaYABBY4A* all grew on SD/-Trp medium, confirming the accuracy of the experimental procedures and operations. pGBKT7-*TaYABBY4A* failed to form monoclonal colonies on SD/-Trp/-Ade/-His/X- α -gal medium, indicating that *TaYABBY4A* does not possess self-activation ability and may require the assistance of other proteins to function. Thus, it can be utilized as a bait to screen for interacting genes in the cDNA library, thereby further exploring the potential functions of *TaYABBY* proteins.

Subcellular localization can reveal the specific cellular compartment in which the *TaYABBY4A* operates. The results (Fig. 6D) indicated that the green fluorescence of the control protein 35S::GFP was observed in the nucleus, cytoplasm, and cell membrane. In contrast, the green fluorescence signal of the 35S::*TaYABBY4A*-GFP was detected solely in the nucleus, with the GFP fluorescence overlapping with DAPI fluorescence. This suggested that the *TaYABBY4A* was localized to the nucleus, aligning with the typical nuclear localization trait of most transcription factors.

Phenotypic characterization of *TaYABBY4A*-overexpressing *Arabidopsis thaliana*

Phenotypic observations were conducted on WT and overexpression lines throughout the growth stage. The results (Figs. 7A-C) indicated that at 25 days, OE-3 and OE-5 exhibited retarded growth and produced fewer rosette leaves. By day 28, WT had entered the flowering stage, whereas OE-3 and OE-5 had not yet

begun bolting. At 50 days, both the number of bolts and the plant height of OE-3 and OE-5 were lower than those of WT. Further statistical analysis of bolting and flowering time (Figs. 7D and 7E) revealed that, compared with the WT, both OE-3 and OE-5 exhibited a significant delay in bolting and flowering. The statistical data on the number rosette leaves indicated that OE-3 and OE-5 had significantly fewer than WT, decreased by 27.03% (OE-3) and 37.84% (OE-5), respectively (Figs. 7F). The diameter of rosette leaves (Figs. 7G) indicated OE-3 and OE-5 had significantly smaller than WT, decreased by 27.46% (OE-3) and 31.15% (OE-5), respectively.

Phenotypic observations of the siliques from WT and overexpression lines indicated that the silique lengths of OE3 and OE5 were significantly shorter than WT, with reductions of 20.30% and 24.87%, respectively (Figs. 8A and 8B). Furthermore, the seed length and diameter of OE3 and OE5 were significantly shorter and narrower than WT. Specifically, the seed length decreased by 26.89% (OE3) and 31.09% (OE5), while the seed diameter decreased by 21.12% (OE3) and 29.58% (OE5) (Figs. 8C and 8D). These findings suggest that the *TaYABBY4A* may be involved in regulating plant growth and development.

Discussion

Genome-wide identification of YABBY gene family in wheat

In the present study, 21 *TaYABBYs* were identified in wheat. There are more YABBY members in wheat than in rice [13], Arabidopsis[20], soybean[10], and tomato[21], indicating that *TaYABBY* has a more complex function. This reason is because wheat is a heterozygous polyploid. Phylogenetic analysis revealed that all *TaYABBYs* are classified into four clades: YAB3, YAB2, INO, and CRC. Among these clades, clade FIL/YAB3 contained the most of *TaYABBYs*, as what was reported in rice [9]. In comparison, *TaYABBYs* were divided into four clades, and no *TaYABBY* protein was found in clade YAB5, which might be lost in the interspecific differentiation process. This phenomenon is also found in *Lactuca sativa*, which loses clade YABBY2[22]. In addition, only four clades (CRC/DL, FIL, INO, and YAB2) were found in rice [13], the YAB5 clade does not exist in rice and other monocots[13], perhaps because YABBY has undergone functional differentiation during the process of plant evolution.

Gene duplication events, particularly tandem duplication and fragment duplication, play an important role in the evolution and functional diversification of genomes. [22, 26, 29]. Wheat experienced multiple gene duplication events in its evolutionary history, including two significant allopolyploidization events. The first polyploidization event involved the combination of the A and B genomes to form tetraploid wheat[30]. The second polyploidization event occurred through hybridization between tetraploid wheat and a D genome donor to form hexaploid wheat[31]. Therefore, we analyzed the duplication patterns of the *TaYABBYs*. There are 21 pairs of fragment duplication genes in *TaYABBYs* that show collinearity. Our results indicate that fragment duplication is the main driving force for the expansion of the *TaYABBYs* during evolution, which is consistent with previous finding in maize [32]. The lack of synteny between wheat and Arabidopsis may be attributed to the fact that Arabidopsis is a dicotyledonous plant, whereas

wheat is a monocotyledonous plant. The early divergence in their evolutionary history has led to significant differences in their genomic structure and sequence. This observation is consistent with the findings of previous studies[8].

The *cis*-acting elements play a crucial role in transcriptional regulation, mediating diverse mechanisms of growth and development[27, 33]. Prior research has demonstrated that *YABBYs* are significantly involved in various biological processes [34, 35]. In this study, a diverse array of plant regulatory *cis*-acting elements were identified within the promoter regions of *TaYABBYs*. Absciscic acid (ABA)-responsiveness and methyl jasmonate (MeJA)-responsiveness elements are critical for mediating plant responses to abiotic stress and disease resistance [29, 35]. Additionally, these two types of elements are extensively involved in fruit development[24]. In wheat, ABA-responsive elements and MeJA-responsive elements are present in each *TaYABBY* (except for *TaYABBY8A/B*), indicating that *TaYABBYs* can respond to various stresses and are involved in seed development. In addition, *YABBYs* can regulate the seed development [24, 36]. In the present study, 18 seed-specific and 20 zein metabolism-regulatory elements were found, suggesting that it may have similar function.

Haplotype analysis can reveal the genetic diversity of wheat and also help us identify signals of domestication and artificial selection [37]. In this study, the haplotype diversity of cultivated varieties was lower than that of landraces and spelt, indicating that these haplotypes may have been under strong selection during wheat domestication. Significant haplotype differences were observed among wheat varieties from different geographical regions, which may be related to local environmental adaptation, domestication history, and artificial selection. The haplotype diversity of spring, winter, and facultative wheat also showed significant differences, which may be associated with their mechanisms of response to temperature and photoperiod. These results indicate that haplotype analysis provides an important molecular basis for wheat genetic improvement.

In this study, based on transcriptome data and qRT-PCR data, we conducted a comprehensive analysis of the expression profiles of *TaYABBYs* across different organs and three distinct endosperm developmental stages. In wheat, *TaYABBY3* is one of the key genes in the FIL/YAB3 clade, and its high expression in the spike may be closely related to spike morphology and the development of floral organs[31]. Consistent with previous studies, the FIL/YAB3 clade demonstrates high expression levels in the spike. Previous studies have indicated that genes within the CRC clade are expressed in reproductive organs, including stigmas and ovules [38]. Notably, in our transcriptome dataset, genes belonging to the CRC clade exhibited elevated expression during the milking stage for each variety, whereas they were expressed at minimal levels or were absent entirely in public databases. This discrepancy may be attributed to the fact that the public databases utilized Chinese Spring as the reference material, whereas our study employed winter wheat. RNA sequencing results revealed variations in the expression levels of *TaYABBYs* throughout the grain-filling stage. These findings suggest that these genes may have direct or indirect impacts on the synthesis and accumulation of starch and protein within the grains [37].

Functional validation of TaYABBY4A overexpression in *Arabidopsis thaliana*

YABBY play multiple roles in plant growth and development. In the CRC subfamily of rice, *OsDL* exhibits high expression in pistils [13]. *Gossypium hirsutum* *GhYABBY6_Dt* is primarily expressed specifically in pistils (carpels) and also shows high expression in cotyledons [39]. Most *YABBY* in *Panicum virgatum*, including *PvYABBY13*, *PvYABBY14*, *PvYABBY15*, and *PvYABBY16*, display relatively high expression levels in seeds and inflorescence tissues [40]. In this study, a combined analysis using public databases and transcriptome data from our laboratory revealed that *TaYABBY4A* is highly expressed in seeds and endosperm, indicating its significant role in the development of reproductive organs. Compared to the WT, overexpression of the *TaYABBY4A* in *Arabidopsis thaliana* resulted in delayed flowering, as well as a reduced number and diameter of rosette leaves. Previous studies have shown that overexpression of wheat *TaYAB1* and soybean *GmFILa* in *Arabidopsis thaliana* also led to delayed flowering [41, 42]. Overexpression of *VvYABBY4* (a YAB2 member from Grapevine) in tomato caused plant dwarfing [14]. These results suggest that *YABBY* gene family is involved in plant growth and development. Overexpression of *VvYABBY4* in tomato significantly reduced the size of fruits and seeds [14]. Overexpression of *BpYAB2*, *BpYAB3*, and *BpYAB4* from *Broussonetia papyrifera* in *Arabidopsis thaliana* also resulted in shorter siliques and smaller seeds [43]. In this study, overexpression of *TaYABBY4A* reduced the length of siliques, as well as the length and diameter of seeds in *Arabidopsis thaliana*, further verifying its role in seed development.

Conclusions

This study systematically explored the *YABBY* gene family in wheat and successfully identified 21 *TaYABBYs*. These genes can be classified into five known subfamilies: FIL/YAB3, YAB2, CRC, INO, and YAB5. The researchers comprehensively analyzed the chromosomal localization, gene structure, evolutionary relationships, and expression patterns of these genes. During the polyploidization process, *TaYABBYs* underwent tandem duplication and gene loss events, accompanied by purifying selection. By integrating RNA-seq and qRT-PCR analysis data, we initially explored the specific expression profiles of *TaYABBYs* in grains. The overexpression of *TaYABBY4A* in *Arabidopsis thaliana* reduced the length of siliques and diameter of seeds. It confirmed that this gene played an important role in plant growth and development as well as grain endosperm development. The research results not only reveal the evolutionary characteristics of the *YABBY* gene family in wheat grains and lay a foundation for polyploid analysis, but also provide a theoretical basis for in-depth exploration of the functions of *TaYABBYs* in grains, as well as a scientific basis for further analysis of *TaYABBYs*.

Material and methods

Plant materials

The Wheat Research Institute at Shanxi Agricultural University supplied the wheat varieties (Yaomai 36, Pinyu 8175, Pinyu 8155, and Yaomai 30 for the experiment. These varieties were planted at the Hancun Experimental Base of the Wheat Research Institute of Shanxi Agricultural University (36°13.2' N, 111°33.7' E) during the 2023–2024 growing season. Each variety occupied a plot measuring 36 meters

in length and 10 meters in width. Wheat ears that blossomed simultaneously were tagged, and grains were harvested at three distinct phases of the grain-filling period: the milk-ripe stage (approximately 10 days post-flowering, MRS), the dough stage (approximately 20 days post-flowering, DP), and the wax-ripe stage (approximately 25 days post-flowering, WRS)[23]. The collected samples were promptly submerged in liquid nitrogen and preserved at -80°C for subsequent RNA-Seq and qRT-PCR analyses.

The wild-type *Arabidopsis thaliana* (Columbia-0, Col-0) was preserved by Professor Nie Xiaojun's laboratory at the state key laboratory for crop stress resistance and high-efficiency production at Northwest A&F University. In the greenhouse of the Wheat Research Institute at Shanxi Agricultural University, wild-type *Arabidopsis thaliana* and overexpression lines were planted in a mixture of black soil, vermiculite, and perlite. Initially, they were grown at 25°C under an 8-hour day/16-hour night photoperiod for 30 days. Subsequently, the growth continued at 25°C under a 16-hour day/8-hour night photoperiod until they reached 60 days. At the 30-day cultivation mark, rosette leaves from various overexpression lines were collected, immediately frozen in liquid nitrogen, and stored at -80°C to determine the expression level of *TaYABBY4A*.

Identification of *YABBYs*

Firstly, the genome information for Chinese Spring (IWGSC v2.1) was downloaded from the JGI website (<https://phytozome-next.jgi.doe.gov/>) by searching for the target species within the Chinese Spring database[44, 45]. The protein sequences for *YABBY* in *Arabidopsis* were obtained from the database TAIR10 (<https://www.arabidopsis.org/>)[44]. The *YABBY* domain (PF04690) was obtained and downloaded from PFAM (<https://pfam.xfam.org/>) for the construction of a Hidden Markov Model (HMM) [46]. *TaYABBYs* were identified with an E-value < 0.01 by using HMMER3.0 [47]. Transcripts from Phytozome (<https://phytozome.jgi.doe.gov/pz/portal.html>) were filtered to retain only the primary ones[48]. TBtools[49] was utilized to extract *TaYABBY* sequences from the Chinese Spring database and to optimize the data. The fundamental physical and chemical properties, such as amino acid length (in base pairs), molecular weight (MW), and isoelectric point (pI), were analyzed using ExPASy-ProtParam (<https://web.expasy.org/protparam/>)[50]. Gene chromosomal locations were determined using Phytozome, and subcellular localization was predicted using Plant-mPLOC (<http://www.csbio.sjtu.edu.cn/bioinf/plant/>)[51].

Conserved domains, phylogenetic tree construction, and chromosomal location

To analyze the evolutionary characteristics of *TaYABBYs* and their phylogenetic relationships among *Arabidopsis*, rice, and wheat, we conducted a series of bioinformatics analyses. First, multiple sequence alignments of *TaYABBYs* were performed using MEGA [64], and the results were visualized with Jalview 2.11.3.2[52] to identify conserved regions. Second, *YABBY* sequences of *Oryza sativa* and *Arabidopsis thaliana* were obtained from PlantTFDB (<http://planttfdb.gao-lab.org/index.php>)[53], and a phylogenetic tree was constructed via the Neighbor-Joining (NJ) method in ClustalW program with a 1000-replication bootstrap value [54], then visualized and classified on iTOL (<https://itol.embl.de>)[55]. Finally, TBtools was

utilized to visualize the chromosomal localization of *TaYABBYs*, facilitating an understanding of their genomic organization.

Gene structure and conserved motif prediction

The gene structures of *TaYABBYs* were analyzed using the GFF annotation file obtained from the wheat Genome Database. Motifs of the *TaYABBYs* were identified using MEME's online tools (<http://meme-suite.org/tools/meme/>). The number of motifs was set to 10. The structure view was drawn using TBtools.[56].

Collinearity analysis

Collinearity analysis of *TaYABBYs* was conducted, and a Circos diagram was generated using the Multiple Collinear Scan Toolkit X (MCScanX) plugin in TBtools[57]. The CDS of *TaYABBYs* were aligned using the BLAST online website(<https://blast.ncbi.nlm.nih.gov/Blast.cgi>)in NCBI[58]. The Ka/Ks Calculator in TBtools computed Ka, Ks, and Ka/Ks ratios for pairs, proteins, and CDS of *YABBYs*[59]. BLAST was used to analyze gene-pair homology. The collinearity among wheat, rice, and Arabidopsis was analyzed and visualized using the Multiple Synteny Plot plug-in of TBtools.

Analysis of cis-acting elements in promoter regions

TBtools software was utilized to obtain the 2000 bp sequence upstream of the CDS of *TaYABBYs*. This sequence was then submitted to Plant-Care (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) for the prediction of promoter *cis*-acting elements[60]. Data processing was conducted using Excel, and visualization was performed with TBtools.

Haplotype analysis

Based on the whole-genome resequencing data of wheat from LuFei[61], SNPs of *TaYABBYs* were extracted in accordance with the chromosomal positions of the genes. The re-sequenced wheat materials were categorized into various haplotypes based on the SNP differences.

Gene expression pattern analysis of wheat

To investigate the expression patterns of *TaYABBYs*, transcriptome data from various tissues and endosperm development stages were obtained from the publicly accessible Triticum aestivum transcriptome database (<http://ipf.sustech.edu.cn/pub/wheatrna/>). Concurrently, grains from four different varieties at the grain-filling stage were submitted to Beijing Tsingke Biotech Co.,Ltd for transcriptome sequencing. The expression patterns of *TaYABBYs* were then visualized on a heatmap using TBtools[62, 63].

cDNA synthesis and quantitative real-time PCR

In gene expression analysis, RNA samples obtained from transcriptome sequencing were used to verify the accuracy of the transcriptome data. For the analysis of expression levels in overexpression lines,

total RNA was extracted from previously collected rosette leaf samples using the Plant Total RNA Extraction Kit (TSINGKE TSP0201). Subsequently, cDNA was synthesized using the TaKaRa reverse transcription kit. Using *TaActin* as the internal reference gene and SYBR Green I as the fluorescent dye, the expression levels in four wheat varieties and ten transgenic lines were quantitatively detected on the Applied Biosystems 7500 Real-Time PCR System (Thermo, Waltham, Massachusetts, USA). All reactions were set up with three technical replicates. The quantitative PCR primers were designed using Primer 5 software, and the specific sequences are detailed in Table S4. The relative gene expression levels were calculated using the $2^{-\Delta\Delta CT}$ method[64].

Vector construction and plant transformation

The full-length coding sequence of *TaYABBY4A* was cloned into the pROKII vector, which contains the CaMV35S promoter. The resulting recombinant plasmid was introduced into the *Agrobacterium tumefaciens* strain EHA105 and subsequently used to transform *Arabidopsis thaliana* (Col-0) via the floral dip method.[65]. The transgenic *Arabidopsis* lines were selected on 1/2 MS medium supplemented with Kanamycin (Phyto Tech, K378, USA, 50 mg/L).

Subcellular localization and transcriptional auto-activation assay

The CDS sequences of *TaYABBY4A*, with their stop codons removed, were cloned using KOD FX Neo (TOYOBO, KFX-201, Japan). The *TaYABBY4A* was then integrated into the N-terminal GFP of the pFGC-eGFP vectors using the *Trelief*TM SoSoo Cloning Kit Ver.2 (Tsingke, TSV-S3, Beijing, China). A single restriction site, *Bam*HI, was used to construct the vector, and the primers are listed in Table S4. The fusion constructs were introduced into onion epidermal cells via particle bombardment (GJ-1000). Subsequently, these constructs were observed under a confocal laser scanning microscope (Zeiss LSM 800, Wetzlar, Germany).

The transcriptional activation activity of *TaYABBY4A* was confirmed using the yeast two-hybrid system. Initially, the CDS of *TaYABBY4A* was amplified (Table S4) to construct the recombinant vector pGBKT7-*TaYABBY4A*. The pGBKT7-*TaYABBY4A*, along with the positive control pGBKT7-PtrWOX13A and the negative control, the empty pGBKT7 vector, were individually transformed into Y2H-Gold yeast cells. The transformed Y2H-Gold yeast cells were then plated onto SD/-Trp (growth control), SD/-Trp/-His/-Ade, and X- α -gal media and cultured at 30°C for 3–5 days to assess their transcriptional activation activity[66].

Phenotypic characterization of overexpression lines

The bolting and flowering times of WT, OE3, and OE5 plants grown for 25, 28, and 50 days were recorded. Furthermore, a ruler was utilized to measure the diameter of the rosette leaves. Siliques and seeds from *Arabidopsis* were randomly chosen, and their lengths, along with the diameter of the seeds, were measured using ImageJ 1.53[67].

Statistical analysis

The experimental data were analyzed using Microsoft Excel 2021 and SPSS 22.0. Significant differences among the various comparisons were determined using Duncan's multiple range test and visualized with GraphPad Prism 10.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of Data and Materials

The genomics information for Chinese Spring wheat (IWGSC v2.1) was obtained from the Joint Genome Institute's website (<https://phytozome-next.jgi.doe.gov/>). The protein sequences of YABBY in rice and Arabidopsis were retrieved from the Plant Transcription Factor Database (<http://planttfdb.gao-lab.org/index.php>). The haplotypes of *TaYABBYs* were extracted based on the chromosomal positions of the genes, using whole-genome re-sequencing data of wheat from Lufei (http://wheat.cau.edu.cn/WheatUnion/b_4/). FPKM values from various wheat organizations were downloaded from the database website (<http://ipf.sustech.edu.cn/pub/wheatrna/>). Data is provided within the manuscript or in supplementary information files.

Competing Interests

The authors declare that they have no competing interests.

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Author's Contributions

Y.J. and X.L. performed the formal data collection and analyses, as well as preparing the original draft of the manuscript. X.M., R.G. participated in the RNA-seq analysis and qRT-PCR analysis. H.J., Y.Z., and X.N. conceived and designed the study, Y.Z., and X.N. obtained funds, and critically reviewed the final draft of the manuscript. All authors have read and agreed to the published version of the manuscript.

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Tables

Tables 1 to 3 are available in the Supplementary Files section.

Figures

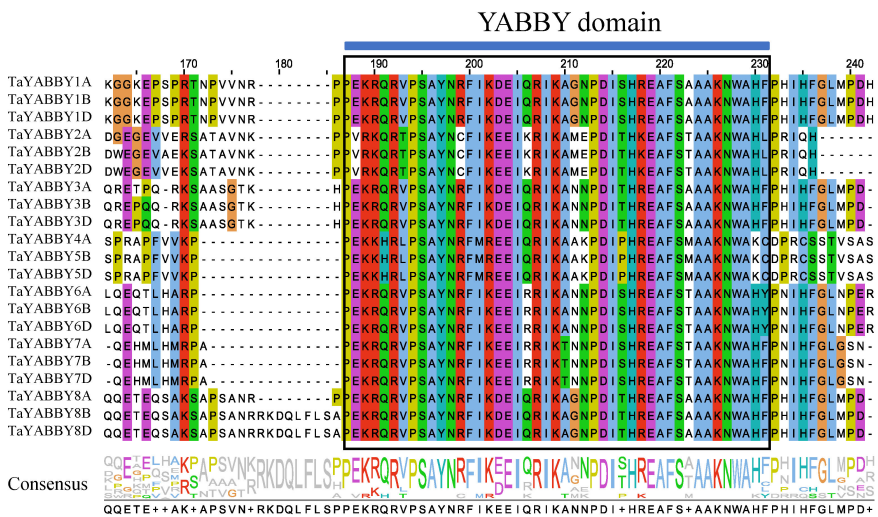
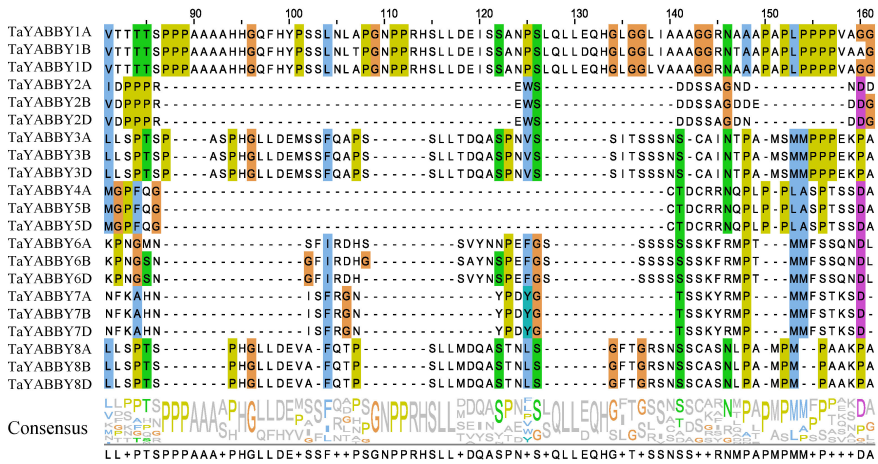
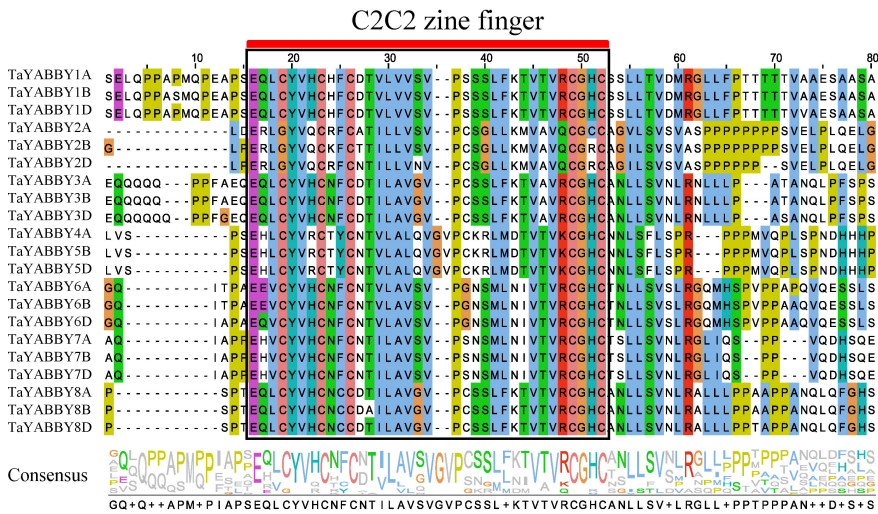


Figure 1

Sequence alignment of the wheat YABBYs. Members of the *TaYABBY* gene family are characterized by two highly conserved domains: a C2-C2 zinc finger domain in the N-terminal and a YABBY domain (PF04690) in the C-terminal.

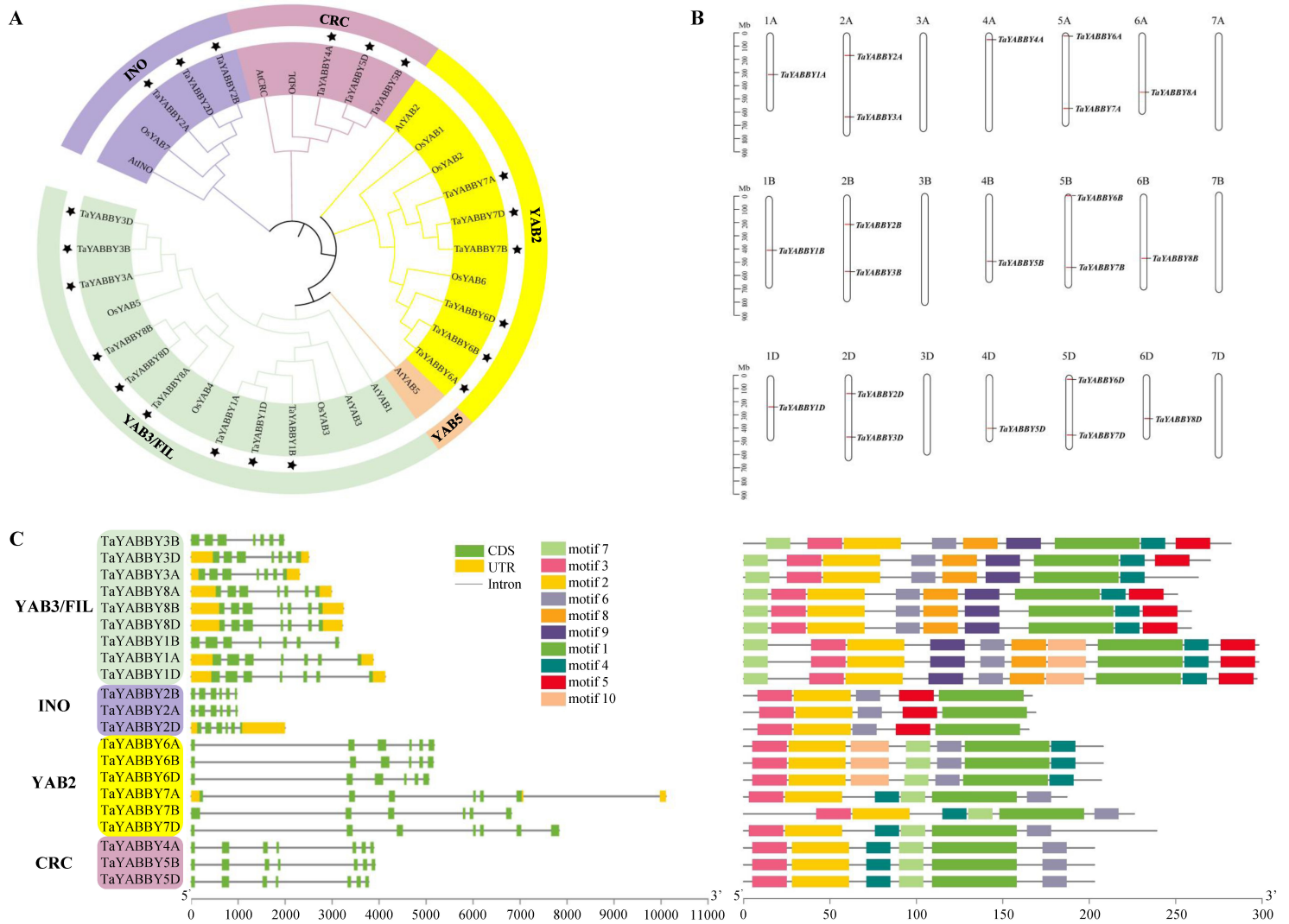


Figure 2

Phylogenetic analysis, chromosomal distribution, gene structure, and conserved motifs of *TaYABBYs*. (A) Phylogenetic relationships among YABBY proteins in *Triticum aestivum* L., *Oryza sativa*, and *Arabidopsis thaliana*. The outer circle is color-coded with light green, orange, yellow, plum red, and mauve, representing the FIL/YAB3, YAB5, YAB2, CRC, and INO subgroups, respectively. Pentacles denote the *TaYABBYs*. ((B) The distribution of *TaYABBYs* on chromosomes. (C) Analysis of the motifs and gene structure of *TaYABBYs*. Exons are depicted as green boxes, UTRs in yellow, and black lines connecting two exons represent introns. Ten conserved motifs are identified in *TaYABBYs*.

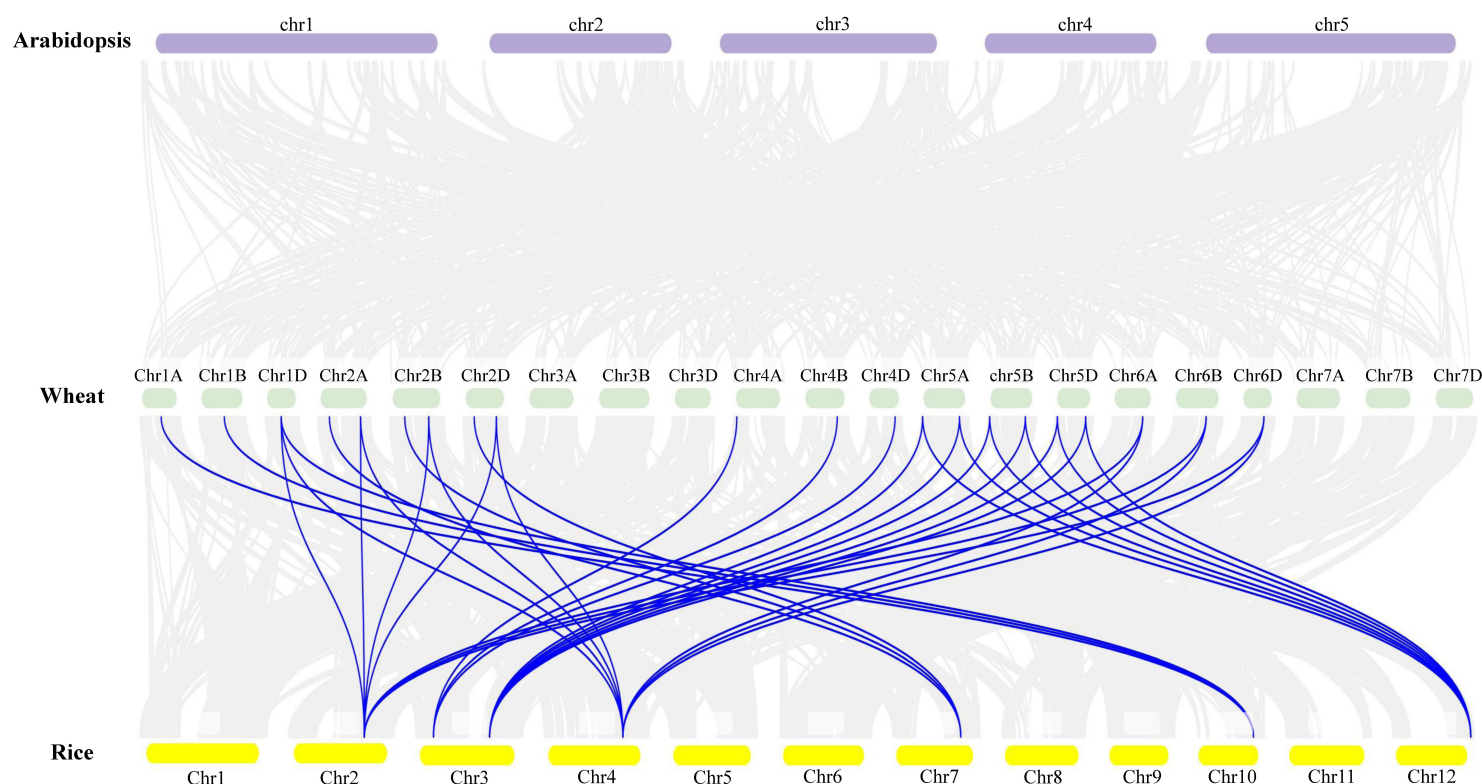


Figure 3

Synteny analysis related to *YABBYs* the genome of Arabidopsis, wheat, and rice. The blue lines represent the colinear *TaYABBYs* pairs.

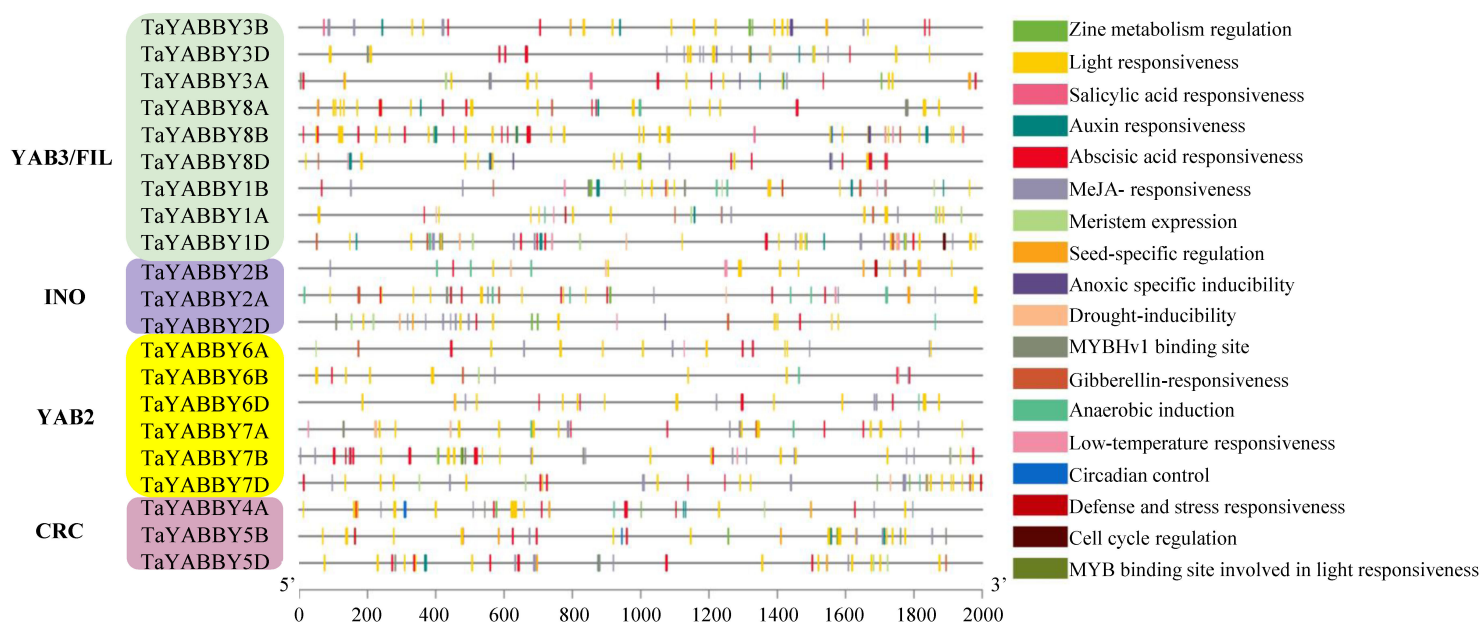


Figure 4

Cis-acting regulatory elements analysis of *TaYABBYs*. Different colors indicate different *cis*-acting regulatory elements.

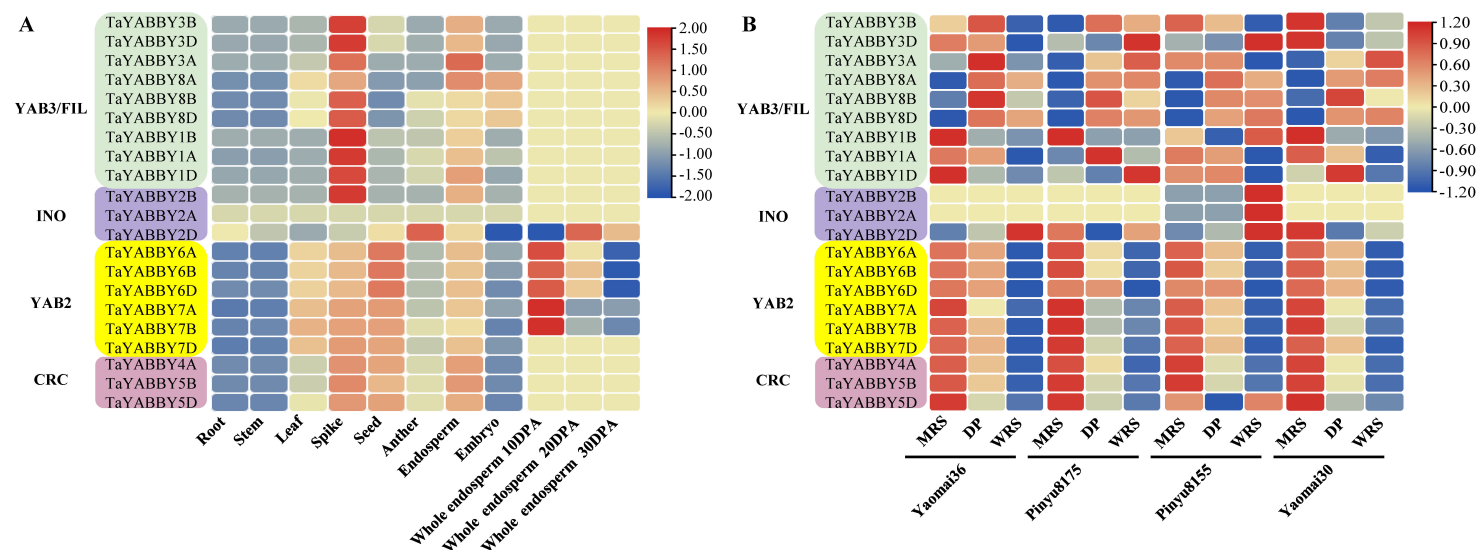


Figure 5

Expression patterns of *TaYABBYs*. (A) Heatmap of expression profiles for *TaYABBYs* in diverse tissues and stages. 10DAP, 20DAP, 30DAP represent the early, middle and late stages of endosperm development, respectively. DPA: Days post-anthesis. (B) RNA-seq analysis of *TaYABBYs* in grain development. Grain expression heat map of Yaomai36, Pinyu8175 and Pinyu8155, Yaomai30, at filling stage. MRS. milk ripe stage, DP. dough period, WRS. wax ripe stage.

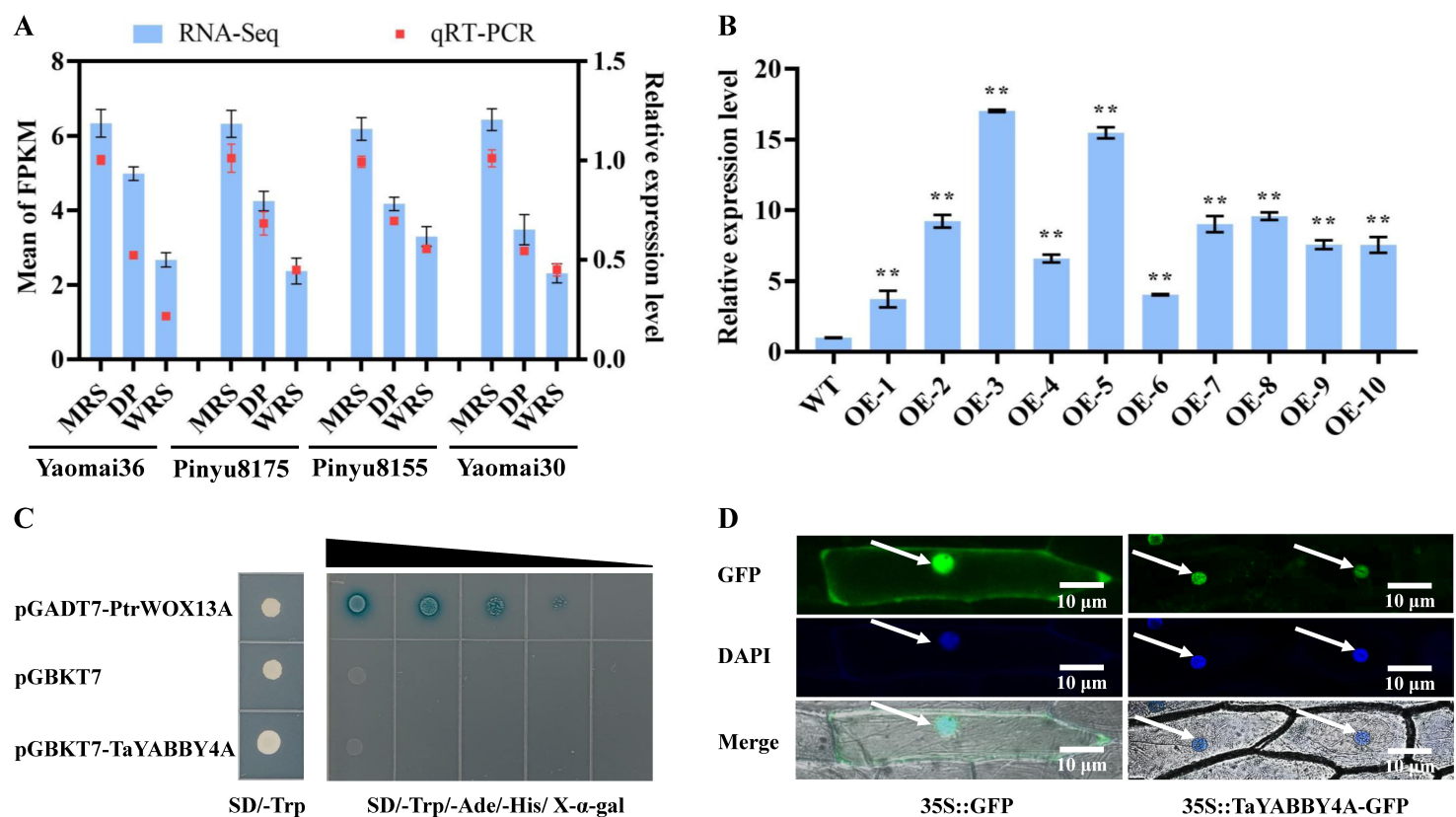


Figure 6

Transcriptome validation and expression analysis by real-time qPCR, subcellular localization, and transcriptional activity of *TaYABBY4A*. (A) *TaActin* was used as a reference gene. Each error bar represents the standard deviation of three biological replicates. MRS. milk ripe stage, DP. dough period, WRS. wax ripe stage. (B) Relative expression analysis of *TaYABBY4A*. Significant differences among different comparisons were determined with Duncan's multiple range test, and significant indicated by ** ($p < 0.01$). (C) Verification of the transcriptional self-activation activity of *TaYABBY4A*. 1, 10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} represent solution dilution ratio of transformed Y2H-Gold yeast cells. (D) Subcellular localization of *TaYABBY4A*. Confocal images manifested the localization of *TaYABBY4A*::GFP in the nuclei of onion epidermal cells. DAPI, a nuclear staining dye; Merge, the merged images of bright-field, GFP, and DAPI staining. 35S::GFP was used as a control. Scale bar = 10 μ m.

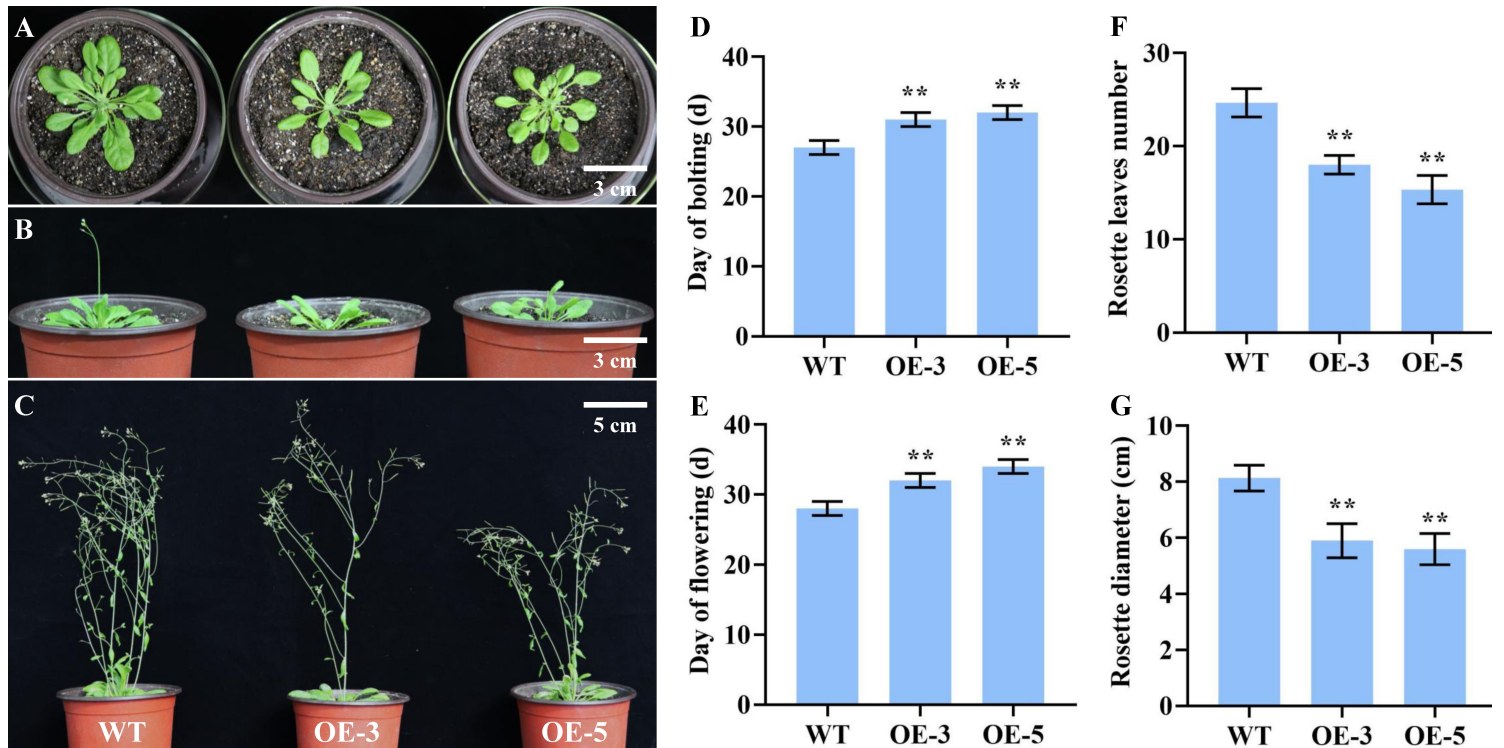


Figure 7

Phenotypic analyses of *TaYABBY4A* transgenic Arabidopsis lines at the growth stage.

(A) Phenotypes of WT and transgenic lines at 25 days, Bar, 3 cm. (B) Phenotypes of WT and transgenic lines at 28 days, Bar, 3 cm. (C) Phenotypes of WT and transgenic lines at 50 days, Bar, 5 cm. (D) Bolting time. (E) Flowering time. (F) Rosette leaf number. (G) Rosette diameter. Duncan's multiple range test was used to determine significant differences between different comparison groups, with ** ($p < 0.01$) indicating significance.

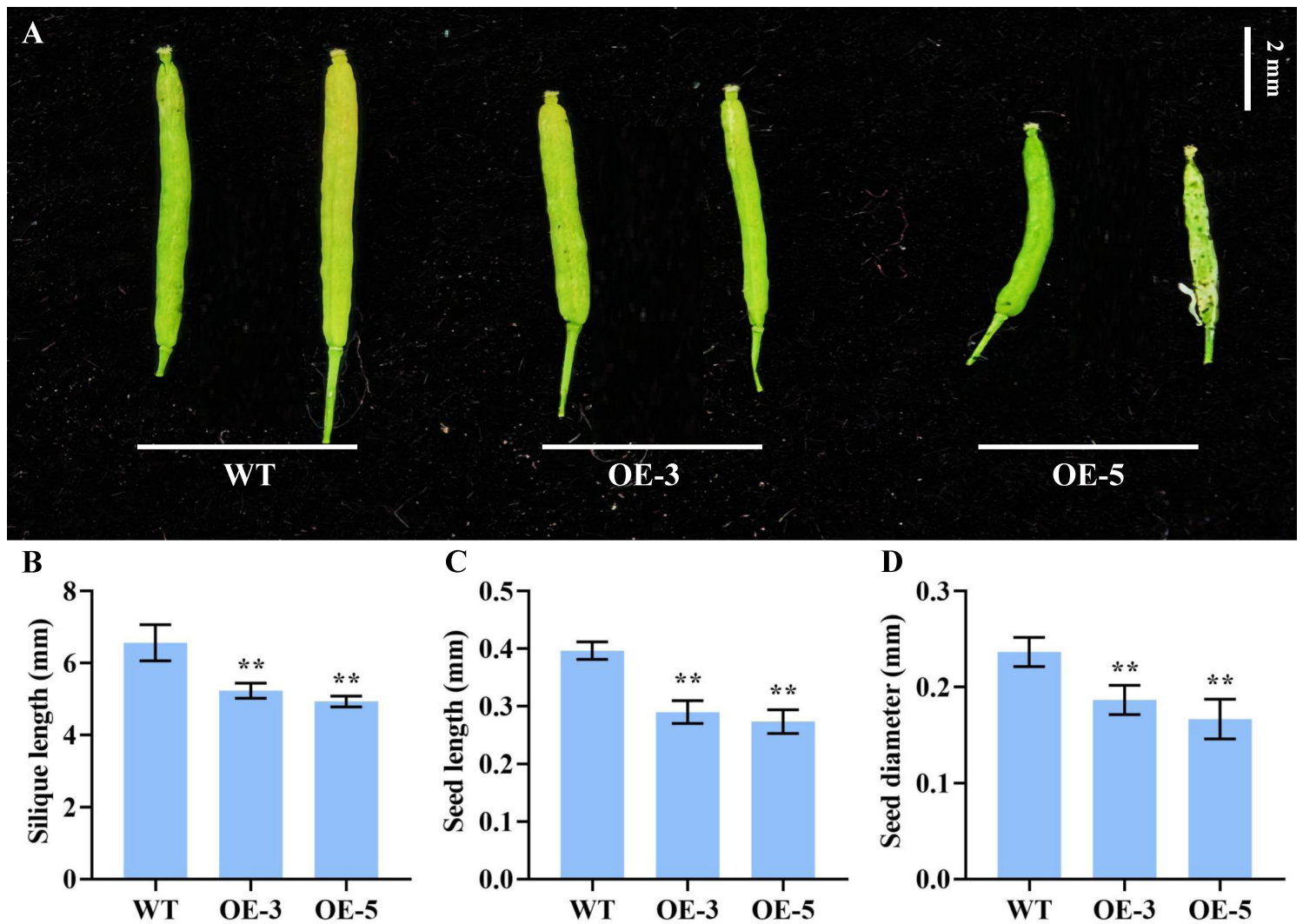


Figure 8

Phenotypic analyses of *TaYABBY4A*in transgenic Arabidopsis siliques and seeds. (A) Siliques of WT and overexpression lines, Bar, 2 mm. (B) Silique length. (C) Seed length. (D) Seed diameter. Duncan's multiple range test was used to determine significant differences between different comparison groups, with **($p < 0.01$) indicating significance.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [Table1.xlsx](#)
- [Table2.xlsx](#)
- [Table3.xlsx](#)
- [SupplementaryMaterials.zip](#)